

INTERNATIONAL GAS TRANSMISSION PIPELINE - QUALITY & INSPECTION PLAN

Pipe Description: 48" (1219 mm) x 18.40 mm, Grade X70M PSL 2, SAWH (Helical Submerged Arc Welded) | **Governing Standard:** API Specification 5L 47th Edition / ISO 3183 PSL 2

No	Inspection / Test Activity	Frequency & Extent	Sampling Location & Specimen	Test Method / Standard	Acceptance Criteria & Specified Limits	Governing Clause	Hold / Witness
1	Ladle Heat Chemical Analysis	One analysis per heat of steel	Ladle sample	ASTM A751	C ≤ 0.12%, P ≤ 0.020%, S ≤ 0.010%, CE_Pcm ≤ 0.23	API 5L 9.2 & Tablo 5	H/W
2	Product Chemical Analysis	Two analyses per heat (different pipes)	Pipe body coupon	ISO 14284	C ≤ 0.12%, P ≤ 0.020%, S ≤ 0.010%, CE_Pcm ≤ 0.23	API 5L 9.2 & Tablo 5	W
3	Pipe Body Tensile Test	Once per test unit (lot of ≤ 100 pipes)	Transverse body strap	ASTM A370	Rt0.5: 485-635 MPa, Rm: 570-760 MPa, Af ≥ 18.5%, Y/T ≤ 0.93	API 5L 9.3 & Tablo 7	H/W
4	Weld Seam Tensile Test	Once per test unit (lot)	Transverse weld specimen	ASTM A370	Rm ≥ 570 MPa across weld seam	API 5L 9.4 & Tablo 7	W
5	CVN Charpy Impact Test (Body) at 0 °C	1 set (3 specimens) per test unit	Transverse body Charpy V	ASTM A370	At 0 °C: Min Average 45 Joules, Min Individual 35 Joules	API 5L 9.8 & Tablo 8	H/W
6	CVN Charpy Impact Test (Weld & HAZ)	1 set weld + 1 set HAZ per test unit	Weld centerline and HAZ	ASTM A370	At 0 °C: Min Average 27 Joules, Min Individual 20 Joules	API 5L 9.8 & Tablo 8	H/W
7	Drop Weight Tear Test (DWTT)	Once per heat (2 specimens) at 0 °C	Transverse DWTT coupon	API RP 5L3	At 0 °C: Average shear fracture area ≥ 85%	API 5L 9.9 & Tablo 18	H/W
8	Guided-Bend Test	1 root + 1 face bend per test unit	Transverse weld coupon	ASTM A370	No defect or crack opening > 3.2 mm in weld/HAZ	API 5L 9.10 & Tablo 18	W
9	Hardness Testing	Once per test unit	Transverse cross section	ASTM E384	Maximum 300 HV10 (Body, HAZ, Weld)	API 5L 10.2.4.8	W
10	Mill Hydrostatic Pressure Test	Each pipe (100% of all pipes)	Full pipe length	API 5L 10.2.6	Test Pressure: 135.5 bar (95% SMYS), Duration: MINIMUM 10 SECONDS	API 5L 10.2.6.2	H/W
11	Ultrasonic Weld Seam Testing (UT)	100% full length of weld seam	Full weld seam	API 5L Annex E	100% automatic UT per ISO 10893-11 acceptance level U2	API 5L Ek E	H/W
12	Pipe Ends UT Laminar Testing	100% of pipe ends (100 mm band)	Both pipe ends	API 5L Annex E.8	100 mm circumferential band on both ends; no defect > 6.0 mm	API 5L Ek E.8	W
13	Diameter and Out-of-Roundness	Once per 4 hours and 100% pipe ends	Pipe body & ends	API 5L 10.2.8.1	Body diameter: +- 0.50% (Max +-4.0 mm); End ovality: per Table 10	API 5L Tablo 10	W
14	Wall Thickness Verification	100% of all pipes (Ultrasonic / Gauge)	Pipe body & ends	API 5L 10.2.8.2	t = 18.40 mm; Tolerance: -8.0% / +15.0% (API 5L Table 11)	API 5L Tablo 11	W
15	Visual Surface Inspection	100% internal and external surface	Full pipe surface	API 5L 10.2.7	No crack, no laminate, no sliver; imperfections ≤ 12.5% t	API 5L 9.12	W
16	Residual Magnetism Measurement	At least once per 4 hours on pipe ends	Pipe ends (4 points)	API 5L 10.2.8.5	Average ≤ 3.0 mT (30 Gauss), Maximum individual ≤ 3.5 mT	API 5L 10.2.8.5	W